King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam II Term 231

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Which one of the following functions can Rolle's Theorem be applied to on the interval $[-1, 1]$.

(a) $f(x) = \dfrac{1}{x^4 + x^2}$

(b) $f(x) = |x| - 1$

(c) $f(x) = 2 + \sin(\pi x)$

(d) $f(x) = |x - 2|$

(e) $f(x) = 2 - x^{2/3}$

2. If $f(t) = t \sin t$, then $f^{(3)}(t) =$

(a) $t^3 \sin^3 t$

(b) $-t \cos t - 3 \sin t$

(c) $t \cos t - \sin t$

(d) $-t \sin t - 3 \cos t$

(e) $2t \cos t - \sin t$

3. If y is defined implicitly as a differentiable function of x by the equation $(x^2 + y^2)^3 = 8xy$, then $\dfrac{dy}{dx} =$

(a) $\dfrac{4y - 3x(x^2 + y^2)^2}{-4x + 3y(x^2 + y^2)^2}$

(b) $\dfrac{8y - 3x(x^2 + y^2)^2}{-8x + 3y(x^2 + y^2)^2}$

(c) $\dfrac{4x - 3y(x^2 + y^2)^2}{-4y + 3x(x^2 + y^2)^2}$

(d) $\dfrac{4y + 3x(x^2 + y^2)^2}{-4x + 3y(x^2 + y^2)^2}$

(e) $\dfrac{4y - 3x(x^2 + y^2)}{-4x - 3y(x^2 + y^2)}$

4. Let $h(4) = 3$, $h'(4) = -\dfrac{1}{4}$, $g(3) = 3$, $g'(3) = -4$. If $f(x) = [g(h(x^2))]^2$, then $f'(2) =$

(a) 24

(b) -8

(c) -6

(d) -24

(e) 12

5. If $f(x) = \dfrac{\cos x}{1 - \sin x}$, then $f'(x) =$

(a) $\dfrac{\sin x}{1 - \sin x}$

(b) $\dfrac{1}{1 - \sin x}$

(c) $\dfrac{1 + \cos x}{(1 - \sin x)^2}$

(d) $\dfrac{1 - \cos x}{1 - \sin x}$

(e) $\dfrac{-\cos x}{(1 - \sin x)^2}$

6. If $y = Ax + B$ is an equation of the tangent line to the graph of $f(x) = (x - 1)e^x$ at the point $(1, 0)$, then $A - B =$

(a) $-2e$

(b) e

(c) 0

(d) $e + 1$

(e) $2e$

7. If $g(x) = \dfrac{x^2}{\sqrt{1 - x^2}}$, then $g'(x) =$

(a) $\dfrac{x}{\sqrt{1 - x^2}}$

(b) $\dfrac{x}{(1 - x^2)^{3/2}}$

(c) $\dfrac{2x - 3x^3}{(1 - x^2)^{3/2}}$

(d) $\dfrac{2x - x^3}{(1 - x^2)^{3/2}}$

(e) $\dfrac{-2x}{\sqrt{1 - x^2}}$

8. A man 2 meters tall walks at a rate of 2.5 meters per second away from a light that is 7 meters above the ground. At what rate is the tip of his shadow moving when he is 5 meters from the base of the light?

(a) 0.7 meter per second

(b) 2.7 meter per second

(c) 0.5 meter per second

(d) 2.5 meter per second

(e) 3.5 meter per second

9. The slope of the tangent line to the graph of

$$f(x) = x \arctan(2x) - \frac{1}{4}\ln(1+4x^2)$$

at the point with x-coordinate $\dfrac{\sqrt{3}}{2}$ is equal to

(a) $\dfrac{\pi}{6}$

(b) $\dfrac{\pi}{3}$

(c) $\dfrac{\pi}{4}$

(d) $\dfrac{2\pi}{3}$

(e) $\dfrac{\pi}{2}$

10. If the length s of each side of an equilateral triangle is increasing at a rate of $5\,\mathrm{m/min}$, then the rate at which the area A of the triangle changes when $s = 4\,\mathrm{m}$ is equal to $\left(A = \dfrac{\sqrt{3}}{4}s^2\right)$

(a) $5\sqrt{3}\ \mathrm{m}^2/\mathrm{min}$

(b) $\dfrac{\sqrt{3}}{5}\ \mathrm{m}^2/\mathrm{min}$

(c) $4\sqrt{3}\ \mathrm{m}^2/\mathrm{min}$

(d) $\dfrac{5\sqrt{3}}{2}\ \mathrm{m}^2/\mathrm{min}$

(e) $10\sqrt{3}\ \mathrm{m}^2/\mathrm{min}$

11. If $y = Ax + B$ is an equation of the normal line to the graph of the equation $y \ln x + y^2 = 0$ at the point $(e, -1)$, then $A^2 + B =$

(a) 0

(b) -1

(c) $-e$

(d) 1

(e) e

12. The sum of the x-coordinates of the points at which the graph of $f(x) = (3x - 2x^2)^3$ has a horizontal tangent line is equal to

(a) $\frac{9}{4}$

(b) 2

(c) 3

(d) $\frac{9}{2}$

(e) $\frac{3}{4}$

13. If $h(x) = \log_3\left(\frac{x\sqrt{x-1}}{2}\right)$, then $h'(2) =$

(a) $\frac{3}{2\ln 3}$

(b) 0

(c) $\frac{2}{\ln 3}$

(d) $-\frac{2}{\ln 3}$

(e) $\frac{1}{\ln 3}$

14. Newton's Method is used to approximate a zero of the function $f(x) = x^3 + x - 3$. If we choose $x_1 = 1$, then $x_2 =$

(a) $\frac{1}{4}$

(b) $\frac{1}{2}$

(c) $\frac{3}{2}$

(d) $\frac{3}{4}$

(e) $\frac{5}{4}$

15. If $y = (\sin x)^{\ln(\cos x)}$, $0 < x < \dfrac{\pi}{2}$, then $y' =$

(a) $(\sin x)^{\ln(\cos x)}\,[\ln(\cos x) - \ln(\sin x)]$

(b) $(\sin x)^{\ln(\cos x)}\,[\cot x \cdot \ln(\cos x) - \tan x \cdot \ln(\sin x)]$

(c) $(\sin x)^{\ln(\cos x)}\,[\tan x \cdot \ln(\cos x) + \cot x \cdot \ln(\sin x)]$

(d) $\ln(\cos x) \cdot (\sin x)^{\ln(\cos x)-1} \cdot \cos x$

(e) $(\sin x)^{\ln(\cos x)}\left[\dfrac{\ln(\cos x)}{\sin x} + \dfrac{\ln(\sin x)}{\cos x}\right]$

16. The sum of the critical numbers of

$$f(x) = x\sqrt[3]{4-x}$$

is equal to

(a) 3

(b) 0

(c) 7

(d) $\dfrac{13}{3}$

(e) $\dfrac{1}{3}$

17. On the closed interval $[-2, 1]$, if

$$f(x) = \frac{6x^2}{x-2}$$

has an absolute maximum at the point (a, b), then $b =$

(a) -1

(b) 10

(c) 48

(d) -6

(e) 0

18. The value of c that satisfies the Mean Value Theorem when applied to $f(x) = \dfrac{x}{x-5}$ on $[1, 4]$ is equal to

(a) $\dfrac{1}{2}$

(b) 2

(c) 3

(d) $\dfrac{1}{3}$

(e) 1

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	Only $2+\sin(\pi x)$ is differentiable on $(-1,1)$ with $f(-1)=f(1)$.
2	**(b)**	$f'=t\cos t+\sin t$, $f''=-t\sin t+2\cos t$, $f'''=-t\cos t-3\sin t$.
3	**(a)**	Differentiate implicitly and solve for y': $\frac{4y-3x(x^2+y^2)^2}{-4x+3y(x^2+y^2)^2}$.
4	**(a)**	Chain rule: $f'(2)=2\,g(3)\,g'(3)\,h'(4)\cdot 4=24$.
5	**(b)**	Quotient rule; numerator $=1-\sin x$, so $f'=\frac{1}{1-\sin x}$.
6	**(e)**	$f'=xe^x$, $A=e$, line $y=ex-e$, so $B=-e$ and $A-B=2e$.
7	**(d)**	Quotient and chain rules: $g'=\frac{2x(1-x^2)+x^3}{(1-x^2)^{3/2}}=\frac{2x-x^3}{(1-x^2)^{3/2}}$.
8	**(e)**	$z=\frac{7}{5}x$, so $\frac{dz}{dt}=\frac{7}{5}(2.5)=3.5$ m/s.
9	**(b)**	$f'(x)=\arctan(2x)$, so the slope is $\arctan\sqrt{3}=\frac{\pi}{3}$.
10	**(e)**	$A'=\frac{\sqrt{3}}{2}s\,s'=\frac{\sqrt{3}}{2}(4)(5)=10\sqrt{3}$ m^2/min.
11	**(b)**	Tangent slope $-\frac{1}{e}$, normal slope e, $B=-e^2-1$, so $A^2+B=-1$.
12	**(a)**	$f'=3(3x-2x^2)^2(3-4x)=0$: $x=0,\frac{3}{2},\frac{3}{4}$; sum $\frac{9}{4}$.
13	**(e)**	Expand the log: $h'=\frac{1}{x\ln 3}+\frac{1}{2(x-1)\ln 3}$, so $h'(2)=\frac{1}{\ln 3}$.
14	**(e)**	$x_2=1-\frac{f(1)}{f'(1)}=1-\frac{-1}{4}=\frac{5}{4}$.
15	**(b)**	Logarithmic differentiation: $y'=y[\cot x\ln(\cos x)-\tan x\ln(\sin x)]$.
16	**(c)**	$f'=\frac{-\frac{4}{3}x+4}{(4-x)^{2/3}}$: critical numbers 3 and 4, sum 7.
17	**(e)**	$f'=\frac{6x(x-4)}{(x-2)^2}$; compare $f(0)=0$, $f(-2)=f(1)=-6$: max value 0.
18	**(c)**	$\frac{-5}{(c-5)^2}=-\frac{5}{4}\Rightarrow c=3$ or 7; only $3\in(1,4)$.